

cell. ③ Once the threshold is crossed, the positive-feedback cycle then rapidly brings the membrane potential close to E_{Na^+} . This stage of the action potential is called the *rising phase*. ④ Two events prevent the membrane potential from actually reaching E_{Na^+} : Voltage-gated sodium channels inactivate soon after opening, halting the flow of Na^+ into the cell, and most voltage-gated potassium channels open, causing a rapid outflow of K^+ . Both events quickly bring the membrane potential back toward E_K . This stage is called the *falling phase*. ⑤ In the final phase of an action potential, called the *undershoot*, the membrane's permeability to K^+ is higher than at rest, so the membrane potential is closer to E_K than it is at the resting potential. The gated potassium channels eventually close, and the membrane potential returns to the resting potential.

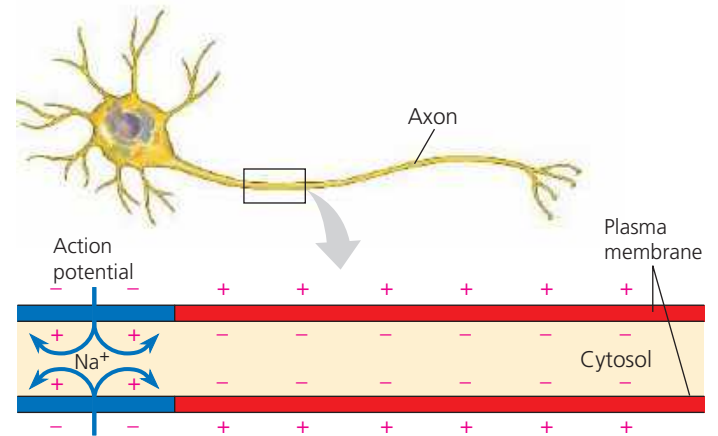
Why is inactivation of channels required during an action potential? Because they are voltage gated, the sodium channels open when the membrane potential reaches the threshold of -55 mV and don't close until the resting potential is restored. They are therefore open throughout the action potential. However, the resting potential cannot be restored unless the flow of Na^+ into the cell stops. This is accomplished by inactivation. The sodium channels remain in the "open" state, but sodium ions cease flowing once inactivation occurs. The end of Na^+ inflow allows K^+ outflow to repolarize the membrane.

The sodium channels remain inactivated during the falling phase and the early part of the undershoot. As a result, if a second depolarizing stimulus occurs during this period, it will be unable to trigger an action potential. The "downtime" when a second action potential cannot be initiated is called the **refractory period**. Note that the refractory period is due to the inactivation of sodium channels, not to a change in the ion concentration gradients across the plasma membrane. The flow of charged particles during an action potential involves far too few ions to change the concentration on either side of the membrane significantly.

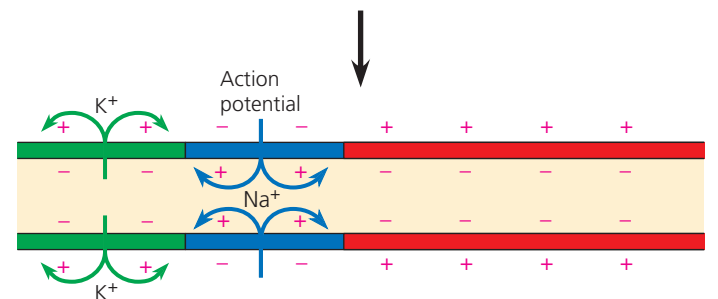
Conduction of Action Potentials

Having described the events of a single action potential, we'll explore next how a series of action potentials moves a signal along an axon. At the site where an action potential is initiated (usually the axon hillock), Na^+ inflow during the rising phase creates an electrical current that depolarizes the neighboring region of the axon membrane (Figure 48.13). The depolarization is large enough to reach threshold, causing an action potential in the neighboring region. This process is repeated many times along the length of the axon. Because an action potential is an all-or-none event, the magnitude and duration of the action potential are the same at each position along the axon. The net result is the movement of a nerve impulse from the cell body to the synaptic terminals, much like the cascade of events triggered by knocking over the first domino in a line.

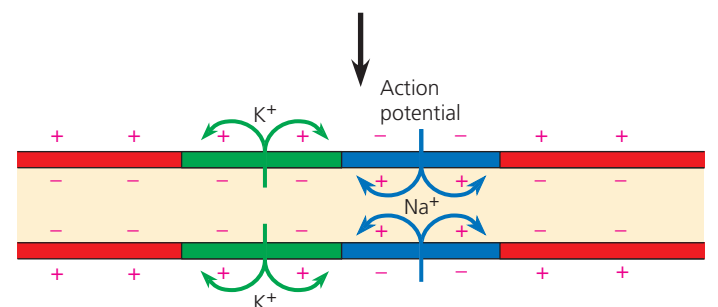
▼ **Figure 48.13 Conduction of an action potential.** This figure shows events at three successive times as an action potential passes from left to right. At each point along the axon, voltage-gated ion channels go through the sequence of changes shown in Figure 48.9. Membrane colors correspond to the action potential phases in that figure.



- 1 An action potential is generated as Na^+ flows inward across the membrane in one region.



- 2 The depolarization of the action potential spreads to the neighboring region of the membrane, reinitiating the action potential there. To the left of this region, the membrane is repolarizing as K^+ flows outward.



- 3 The depolarization-repolarization process is repeated in the next region of the membrane. In this way, local currents of ions across the plasma membrane cause the action potential to be propagated along the length of the axon.

DRAW IT For the axon segment shown, consider a point at the left end, a point in the middle, and a point at the right end. Draw a graph for each point showing the change in membrane potential over time at that point as a single nerve impulse moves from left to right across the segment.

➔ **Mastering Biology BioFlix® Animation: Conduction of an Action Potential**

An action potential that starts at the axon hillock moves along the axon only toward the synaptic terminals. Why? Immediately behind the traveling zone of depolarization, the